PPF Introduction

▼ **How can the production possibility frontier model illustrate the economic problem?**

The PPF is a model that is useful for understanding certain aspects of the economic problem and relative scarcity

An economy's PPF shows the possible output levels of two goods that can be produced using the economy's current resources and factors of production.

The PPF therefore shows the maximum amount of one product that can be produced for each given level of output of another product. Each PPF is drawn on the assumption that resources and technology's
are fixed and there are only two goods produced.

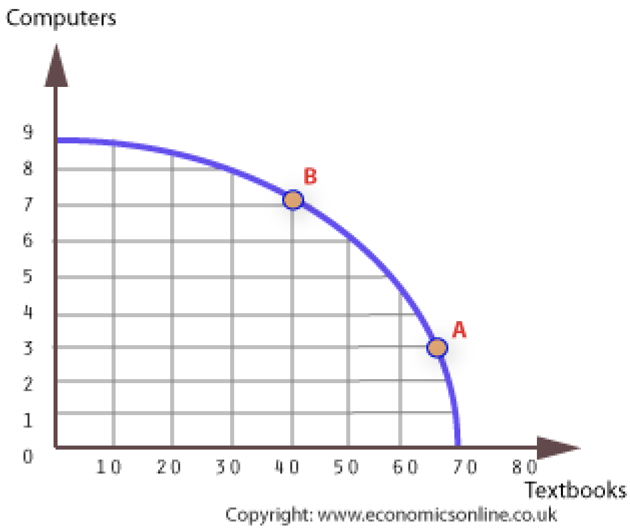
The model helps show the opportunity cost of moving from one of the production combinations to another. It also shows potential output. The level of actual output will be somewhere inside the frontier if the economy is not operating at full capacity.

▼ **When will the PPF be a straight line and when will it be a curve?**

If all the economy's resources are equally productive when being used to make each product the PPF will be a straight line. Doubling the resources being used to make consumption goods will double the output of consumption goods.

However in many cases the PPF will not be a straight line. This is because of the Law of Diminishing Marginal Returns. This law states that, given a fixed amount of other resources to work with, the output of a new worker, for example, will be less than the output of the previous workers, because adding extra workers dilutes the amount of capital equipment available to each worker and the new workers may not have appropriate skills.

▼ **How do we interpret PPFs?**



We can describe the opportunity cost to this company as a given output of computers or textbooks.

For example, the opportunity cost of producing 3 million computers is 5 million textbooks. This is calculated because at point A (3m computers and 65m textbooks) you are missing out on the maximum output of textbooks which is 70 million, therefore, 70m - 65m is 5m which is the opportunity cost of producing 3 million computers in terms of textbooks. Similarly, the opportunity cost of producing 7m computers (point B) is 31m textbooks (70m-39m)

▼ **What is Pareto efficiency?**

Points on the PPF such as A and B are said to be efficient and indicate that an economy's scarce resources are being fully employed. This is called Pareto efficiency.

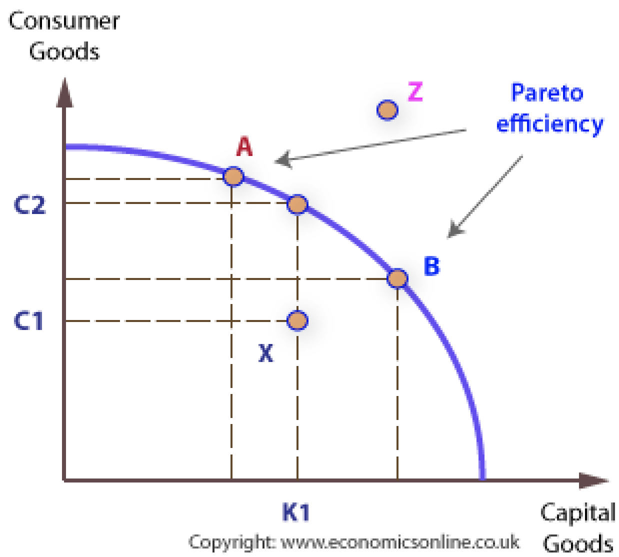
Any point which is inside the frontier such as X is said to be inefficient and not all resources are being used and the output could be greater.

Points such as Z are said to be impossible with the economies current resources but may be a future objective.

Pareto efficiency is unlikely to occur in the real world because of rigidities and imperfections. For example, it is unlikely that all resources can be fully employed at all times because some workers may be in the process of training or finding new jobs. Similarly, an entrepreneur may have wound up one business venture and be in the process of setting up another one. In the mean time he is unproductive to the economy.

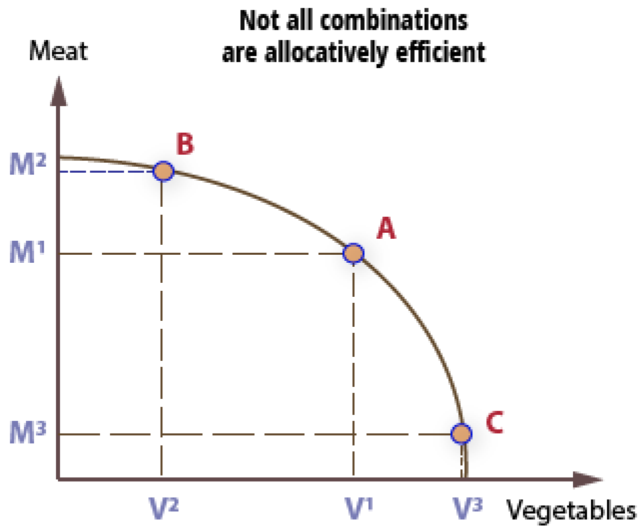
However, this concept is useful for two reasons:

1. It can be an objective for the economy to direct itself towards
2. It can help highlight imperfections and rigidities in the existing economy that prevent Pareto efficiency being achieved



▼ Productive and Allocative Efficiency

Any point on the frontier of the PPF is by definition employing all of the economy's resources in an efficient way and there is no waste or unemployment. However from the consumers point of view a particular combination of goods may not be allocatively efficient. For it to be allocatively efficient it must satisfy all consumer demands and preferences. Allocative efficiency is more formally expressed as a level of output where the marginal benefit to the consumer or, the last unit consumed equals the marginal cost of supply of that unit. And not all combinations of goods will satisfy this condition.

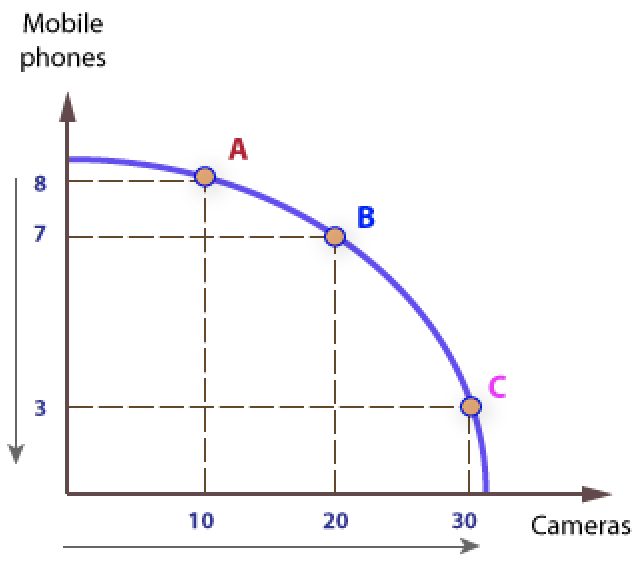


In the above example, a society may only produce meat or vegetables. If its population prefers a varied diet such as point A, points B and C are less allocatively efficient because they do not meet society's preferences.

▼ Increasing Opportunity Cost

Opportunity cost can be thought of as how a decision to increase the production of one good leads to a decrease in production of another good.

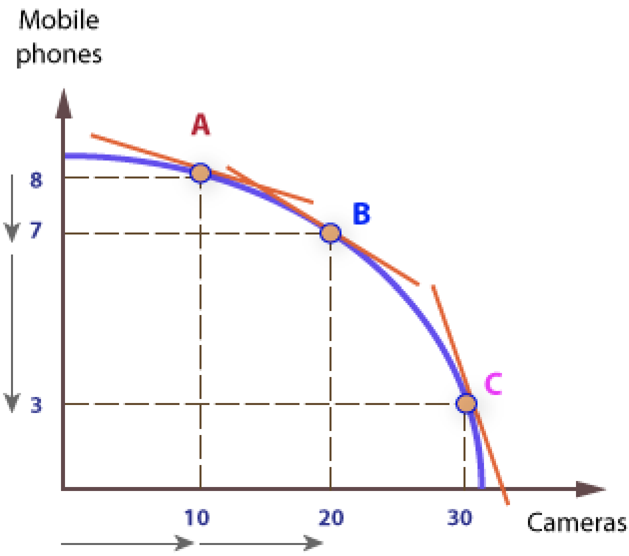
According to economic theory, successive increases in the production of one food will lead to an increasing sacrifice in terms of a reduction in the other goods production. For example, as an economy tries to increase the production of good X, such as cameras, it must sacrifice more of the other good, Y, such as mobile phones.



This is one reason why the PPF is concave. If an economy initially produces at A, with 8m phones and 10m cameras, and then increases output of cameras by 10m to 20m, it must sacrifice 1m phones as it moves to point B.

If it now wishes to increase output of cameras by a further 10m to 30m it must sacrifice 4m phones rather than 1m as the economy moves to point C; hence, opportunity cost increases as more of the good is produced.

The gradient of the PPF gets steeper as more cameras are produced, indicating a greater sacrifice in terms of mobile phones foregone.


Mobile phones
A
B
C
8
7
3
10
20
30
Cameras